# Impact energy and TNT equivalents

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Convert an energy estimate without claiming a complete damage prediction.

**SUCCESS CRITERIA - I can...**

1. Convert speed units before squaring.
2. Calculate kinetic energy from stated model values.
3. Distinguish an energy equivalent from an explosion or damage forecast.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Bring back an earlier idea

A constant 30 N forward force acts at steady 5 m/s for 7 s. Find mechanical power and work.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $K = \frac{1}{2}mv^2$
TNT equivalent (tons) = $K/(4.184\times10^9\ J)$.
1 ton TNT equivalent = $4.184\times10^9$ J. Equal energy does not imply identical effects.

**Worked reasoning:** A fictional event transfers $8.4\times10^6$ J. Using $4.2\times10^6$ J per kg TNT as an energy comparison gives 2 kg TNT equivalent. This compares energy, not identical effects.

**Your turn:** Draw a labeled energy comparison and name one effect this number alone cannot predict.

**Marzano check - circle:** ① ② ③

## 03 Represent and interpret

A report gives impact energy in J and another in TNT-equivalent tons. Are they necessarily describing different physical quantities?

**Marzano check - circle:** ① ② ③

## 04 Solve and justify

Hypothetical mass $10^6$ kg at 10 km/s. Calculate K and TNT equivalent tons.

**Marzano check - circle:** ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

### 05 Analyze the reasoning

Why should an energy-equivalent number not be reported as a precise area of destruction?

**Marzano check - circle:** 1 2 3

### 06 Choose a method

A 5 kg object descends 7 m near Earth. Use g=10. Find ΔU and the work done by gravity.

**Marzano check - circle:** 1 2 3

### 07 Exit check

At fixed speed, mass doubles while assumed energy coupling fraction halves. Compare coupled energy.

**Marzano check - circle:** 1 2 3

### 08 STRETCH: Apply it another way

A 2000 kg hypothetical object moves at 1000 m/s. Find kinetic energy and kg TNT equivalent using $4.2\times10^{6}$ J/kg. Explain why neither alone predicts a crater size.

**Marzano check - circle:** 1 2 3